Universiti Tun Hussein Onn Malaysia

UNIVERSITI TUN HUSSEIN ONN MALAYSIA

FINAL EXAMINATION
SEMESTER I
SESSION 2024/2025

COURSE NAME	:	DATABASE
COURSE CODE	:	BIC 21404
PROGRAMME CODE	:	BIS / BIP
EXAMINATION DATE	:	FEBRUARY 2024
DURATION	:	3 HOURS
INSTRUCTIONS	:	1. ANSWER ALL QUESTIONS 2. THIS FINAL EXAMINATION IS CONDUCTED VIA ☐ Open book ☒ Closed book 3. STUDENTS ARE **PROHIBITED** TO CONSULT THEIR OWN MATERIAL OR ANY EXTERNAL RESOURCES DURING THE EXAMINATION CONDUCTED VIA CLOSED BOOK

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THIS QUESTION PAPER CONSISTS OF **FOUR (4)** PAGES

Q1 Questions **Q1(a)** to **Q1(e)** are based on **Table Q1.1** and **Table Q1.2**. **Table Q1.1** and **Table Q1.2** represents a hypothetical list of `Employees` and `Department` in a company respectively.

Table Q1.1 Employees

id	name	position	salary	Depart_id
s101	Ali	Manager	80,000.00	d101
s102	Nur Anisa	Analyst	50,000.00	d101
s103	Muaz	Analyst	55,000.00	d101
s104	Emily	Manager	85,000.00	d102
s105	Fatimah	Analyst	50,000.00	d102
s106	Kassim	Programmer	10,000.00	d104
s107	Abu Bakar	Analyst	45,000.00	d101
s108	Junaida	Programmer	10,000.00	d101

Table Q1.2 Department

depart_id	Depart_name	Manager_id	location
d101	Software	Manager	Kuala Lumpur
d102	Network	Analyst	Serdang
d103	Admin	Analyst	Kuala Lumpur
d104	Technical	Manager	Serdang

(a) Write a SQL statement to update employee with the following details:

`ID = 's108', name = 'Aidil', position = 'Analyst', salary = 15,000.00 and depart = 'Software'.`

(6 marks)

(b) Suppose Emily has been promoted to a new position as Senior Manager and her salary has been increased to 95,000.00. Write an SQL statement to update Emily's record with the new `position` and `salary`.

(6 marks)

(c) Write a SQL statement to count how many positions exist in the company.

(2 marks)

(d) Write a SQL statement to display name, salary from the smaller to highest and their location.

(6 marks)

(e) Explain referential integrity with an example from **Table Q1.1** and **Table Q1.2**.

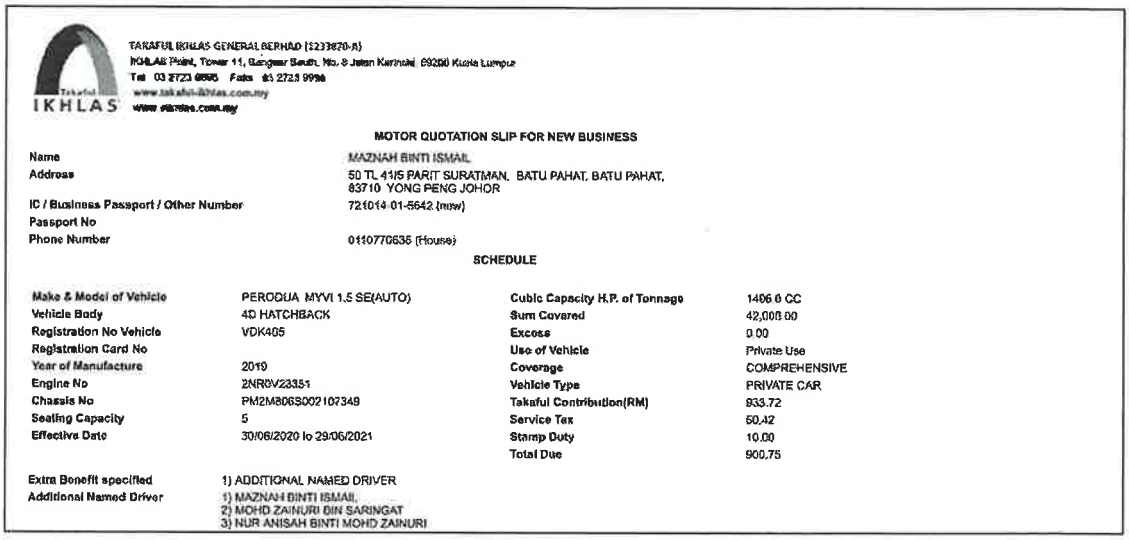
(4 marks)

(f) Write a SQL statement to create table Employees with referential integrity to table Department.

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(6 marks)

Q2 Answer **Q2** based on **Figure Q2.1.**



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MOTOR QUOTATION SLIP FOR NEW BUSINESS

Name	MAZNAH BINTI ISMAIL
Address	50 TL 41/5 PARIT SURATMAN, BATU PAHAT, BATU PAHAT, 83710 YONG PENG JOHOR
IC / Business Passport / Other Number	721014 01 5642 (new)
Passport No	
Phone Number	0110770535 (House)

SCHEDULE

Make & Model of Vehicle	PERODUA MYVI 1.5 SE(AUTO)	Cubic Capacity H.P. of Tonnage	1496.0 CC
Vehicle Body	4D HATCHBACK	Sum Covered	42,000.00
Registration No Vehicle	VDK405	Excess	0.00
Registration Card No		Use of Vehicle	Private Use
Year of Manufacture	2019	Coverage	COMPREHENSIVE
Engine No	2NR0V23351	Vehicle Type	PRIVATE CAR
Chassis No	PM2M806S002107349	Takaful Contribution(RM)	933.72
Seating Capacity	5	Service Tax	50.42
Effective Date	30/06/2020 to 29/06/2021	Stamp Duty	10.00
		Total Due	900.75

Extra Benefit specified	1) ADDITIONAL NAMED DRIVER
Additional Named Driver	1) MAZNAH BINTI ISMAIL 2) MOHD ZAINURI BIN SARINGAT 3) NUR ANISAH BINTI MOHD ZAINURI

Figure Q2.1

(a) Give **FOUR (4)** entity.

(4 marks)

(b) Draw an entity-relationship diagram (ERD). All entities must include with suitable attributes, primary key and relationship.

(16 marks)

Q3 Table **Q3** shows information on student's subjects' registration.

Table Q3

studId	name	address	Post-code	state	subjectId	subjectName	marks	grade
AI1010	Awie	No 10	83000	Johor	BIC20104	Database	89	A
					UMS10101	English	65	C
					BIE10101	Java	78	B
AI1210	Azie	13 Lorong 1	41000	Selangor	BIC20104	Database	98	A
					BIS10111	Networking	68	C
CI1520	Muaz	Kg New	27000	Pahang	UMS10101	English	80	A
					BIS10111	Networking	68	C
					BIE10101	Java	78	B

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(a) Discuss the differences between First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF)

(4 marks)

(b) Construct normalize table based on **Table Q3.**

(i) From UNF to 1NF.

(4 marks)

(ii) From 1NF to 2NF.

(4 marks)

(iii) From 2NF to 3NF.

(4 marks)

Q4 Answer Q4(a) to Q4(c).

(a) How relevant is the data warehouse in now days.

(4 marks)

(b) Describes the impact of database design based on structure data and unstructured data.

(10 marks)

- END OF QUESTIONS -

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